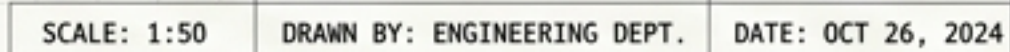


From Physical Iron to Virtual Agility



MIGRATION IS A BUSINESS CONTINUITY PROJECT, NOT A FILE TRANSFER

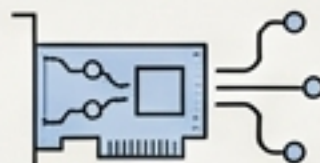
TECHNICAL CONVERSION



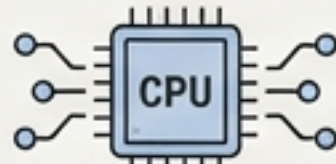
1. Hypervisors



2. Virtual Disks



3. Virtual Network Interfaces



4. CPU Translation

BUSINESS CONTINUITY



1. Application Availability



2. Data Integrity



3. Operational Reliability



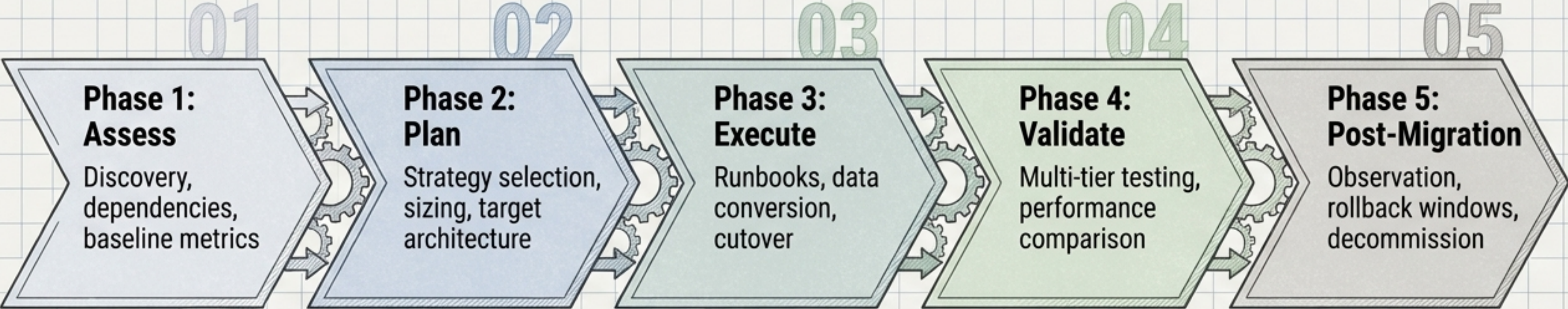
4. Security Posture

A VM that boots but cannot serve users is a failed migration.



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The 5-Phase Professional Migration Lifecycle



Skipping stages creates **operational debt**.
Every technical decision lives inside this blueprint.

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Phase 1: Assessment – The Illusion of Installed Specs

The Trap

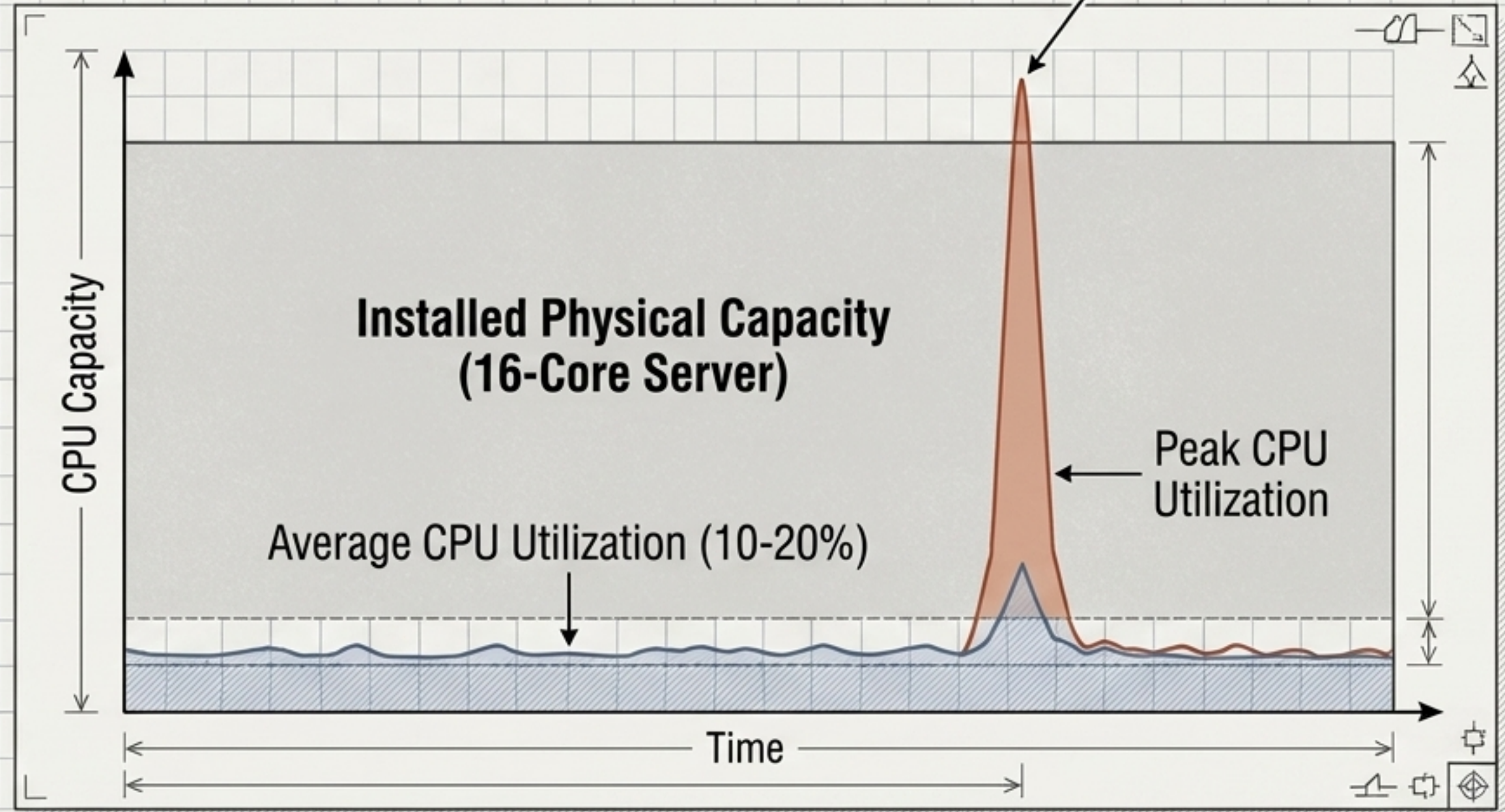
Copying physical hardware specs wastes licensing and hardware budget.

The Reality

Sizing must be based on actual utilization measurements, tracking peak duration and load patterns.

The Rule

Design for the peak, not the average. Over-provisioning wastes money; under-provisioning kills performance.



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Phase 1: Mapping the Dependency Web

The Mistake

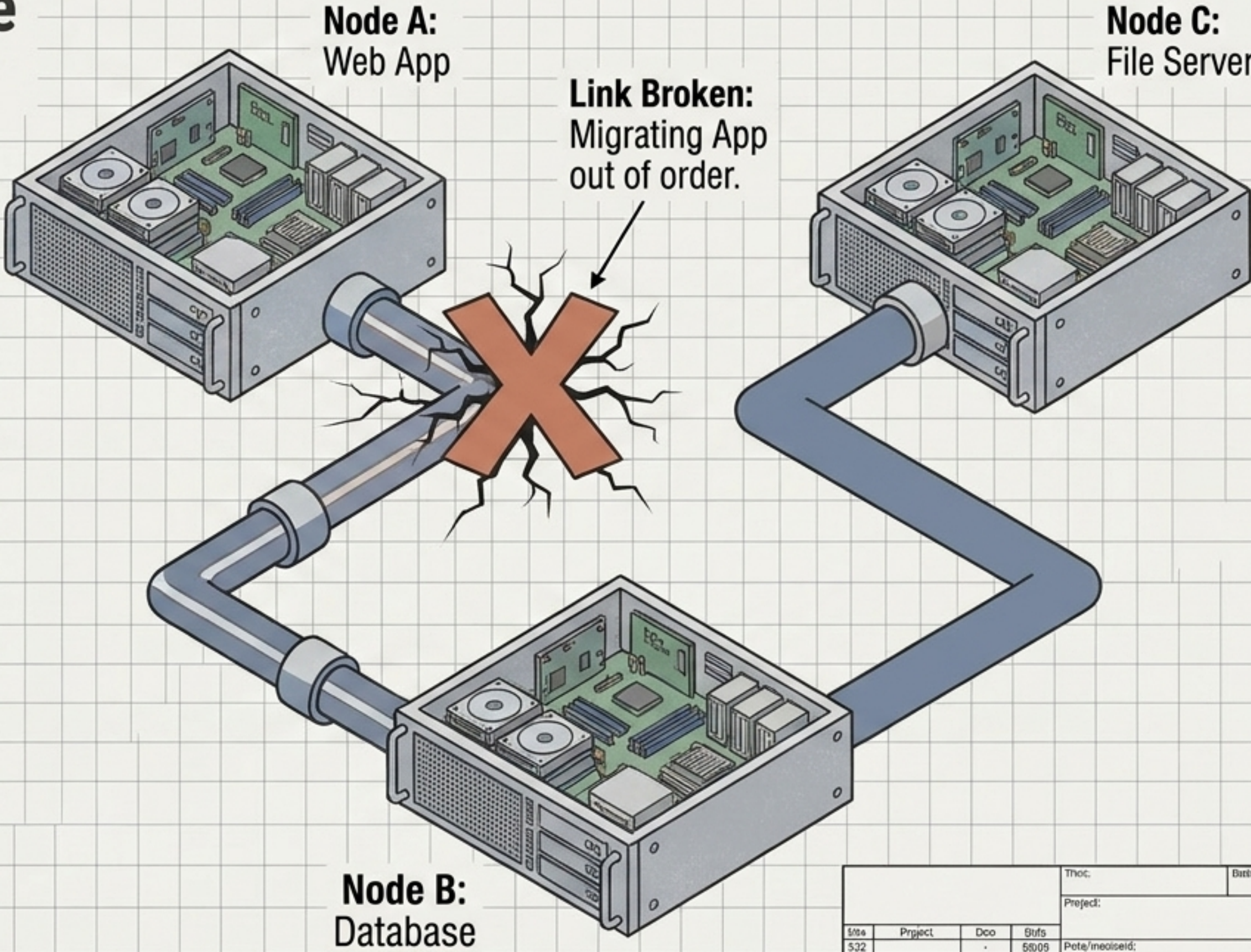
Migrating servers alphabetically or by rack unit.

The Impact

If the Application VM is migrated before the Database, legacy APIs break, latency spikes, and services halt.

Discovery Methods

Discover dependencies through firewall logs, network flow data, DNS records, and owner interviews—not just outdated documentation.



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Phase 2: The Strategy Selection Matrix

Physical-to-Virtual (P2V)	The Rebuild (App Migration)	Virtual-to-Virtual (V2V)
Pros: – Fast execution – Preserves application state Cons: – Imports legacy junk – Old drivers and OS rot – Old drivers and OS rot	Pros: – Clean environment – Modernizes OS – Fixes configuration drift Cons: – Requires fresh OS installation – Data migration time	Use Case: – Platform shifts (e.g., KVM) Complications: – Virtual disk formats – Hypervisor translation – Network adapter changes

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Phase 2: Execution Modalities vs. Downtime

High Downtime

Zero Downtime

Cold Migration

Workload stopped before moving. Simple, consistent, high downtime.

Warm Migration

Data copied while active, followed by a short sync/cutover. Low downtime.

Live Migration

VM moved between compatible hosts while running. Zero noticeable downtime.

Replication-Based

Continuous sync to target. Minimal downtime, but requires application-aware consistency for databases.

Define RTO (Recovery Time) and RPO (Recovery Point) before selecting a tool.

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Phase 2: Target Architecture & The Overcommitment Trap

Safe if workloads do not all demand maximum CPU simultaneously.



CPU Overcommitment

Memory pressure causes severe performance degradation, swapping, and ballooning.



Memory Overcommitment

Overcommitment is a capacity-planning decision, not free performance.

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Phase 3: The Crucial Distinction — Backup vs. Rollback

BACKUP



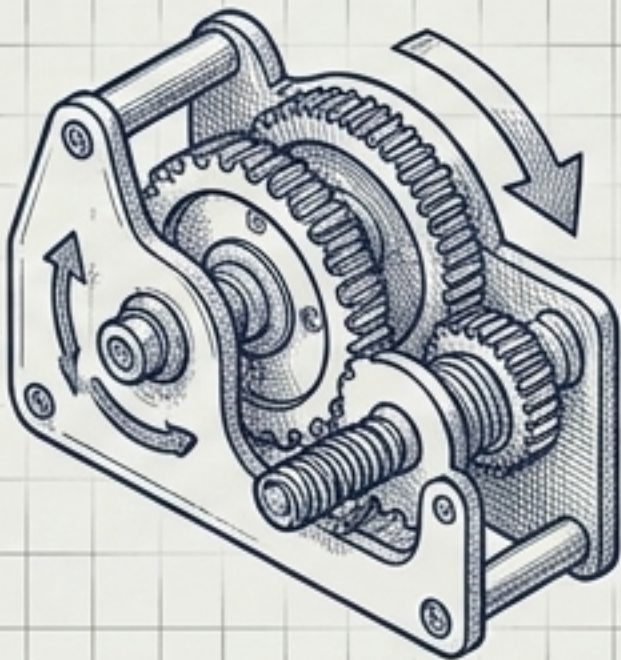
Definition: A recoverable copy of data.

Function: Protects against data corruption or loss.

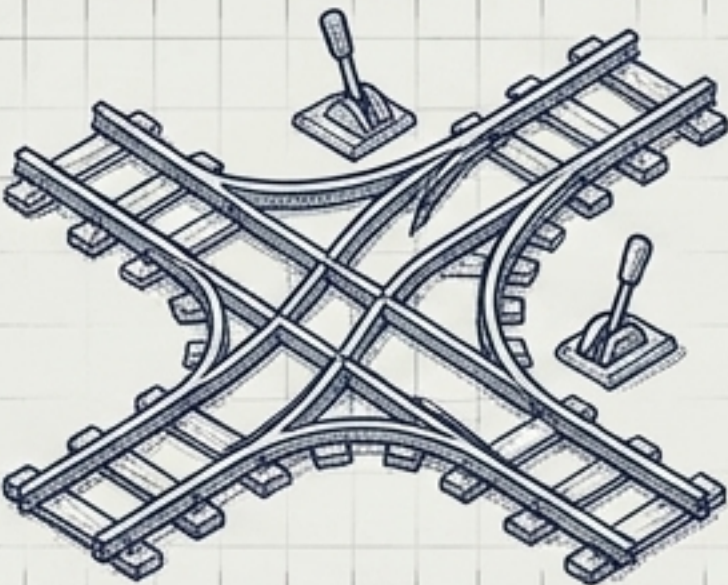
Definition:
A recoverable copy of data.

Limitation:
Does not guarantee an immediate operational return to service.

ROLLBACK



Function:
The emergency ripcord for a failed change.



Requirement:
Must be a predefined, tested procedure.

Never start a migration without a tested rollback plan.

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Phase 3: Anatomy of a Migration Runbook

Amateur Runbook

[] Step 4: Check Network.

Professional Runbook

- [] Step 4.1: Verify DNS propagation for 10.10.10.20.
- [] Step 4.1: Verify DNS propagation for 10.10.10.20.
- [] Step 4.2: Confirm port 443 listening on web server.
- [] Step 4.3: Ping database gateway with <1ms latency.

A runbook is a detailed operational procedure moving the team from improvisation to disciplined execution. It must be specific, measurable, and executable by any qualified engineer, not just the author.

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Phase 4: The 5-Level Validation Pyramid

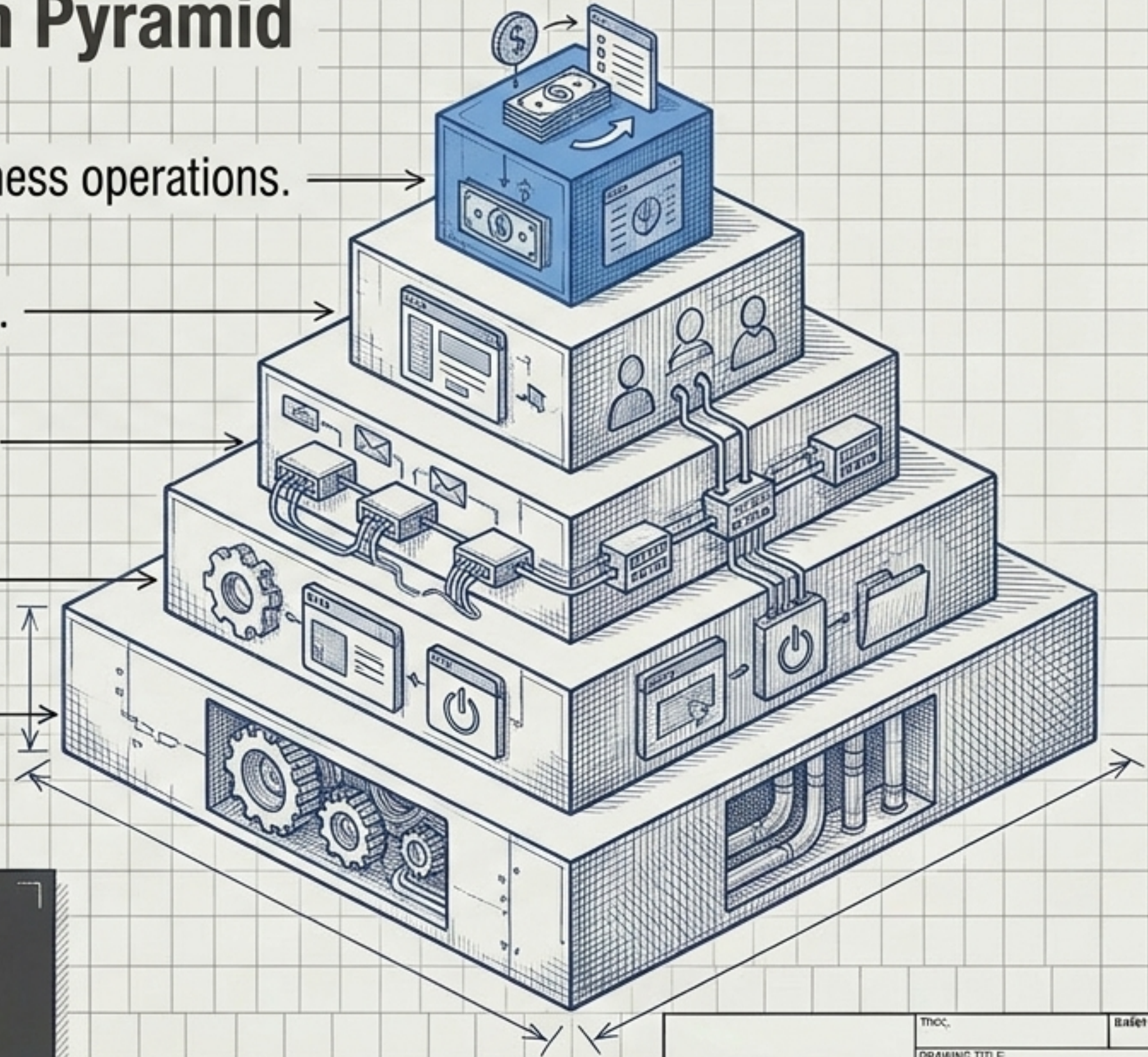
5. **Business:** Users can transact and perform business operations.

4. **Application:** The application interface functions.

3. **Network:** Required systems can communicate.

2. **Operating System:** Critical services start.

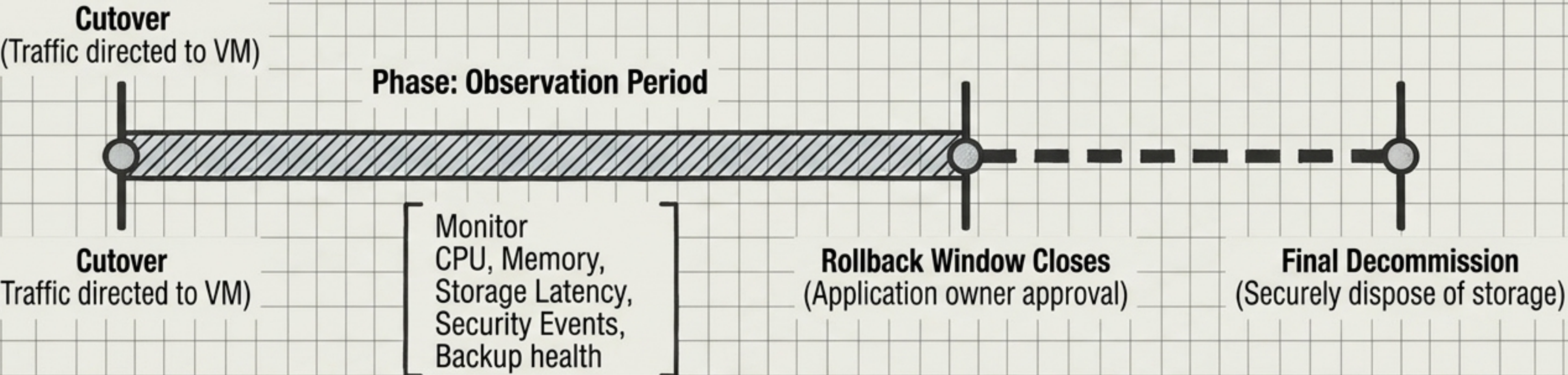
1. **Infrastructure:** The VM powers on.



Do not declare victory at Level 1.
A VM that boots perfectly while the
application is broken is a migration
failure.

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Phase 5: The Post-Migration Trap & Decommissioning



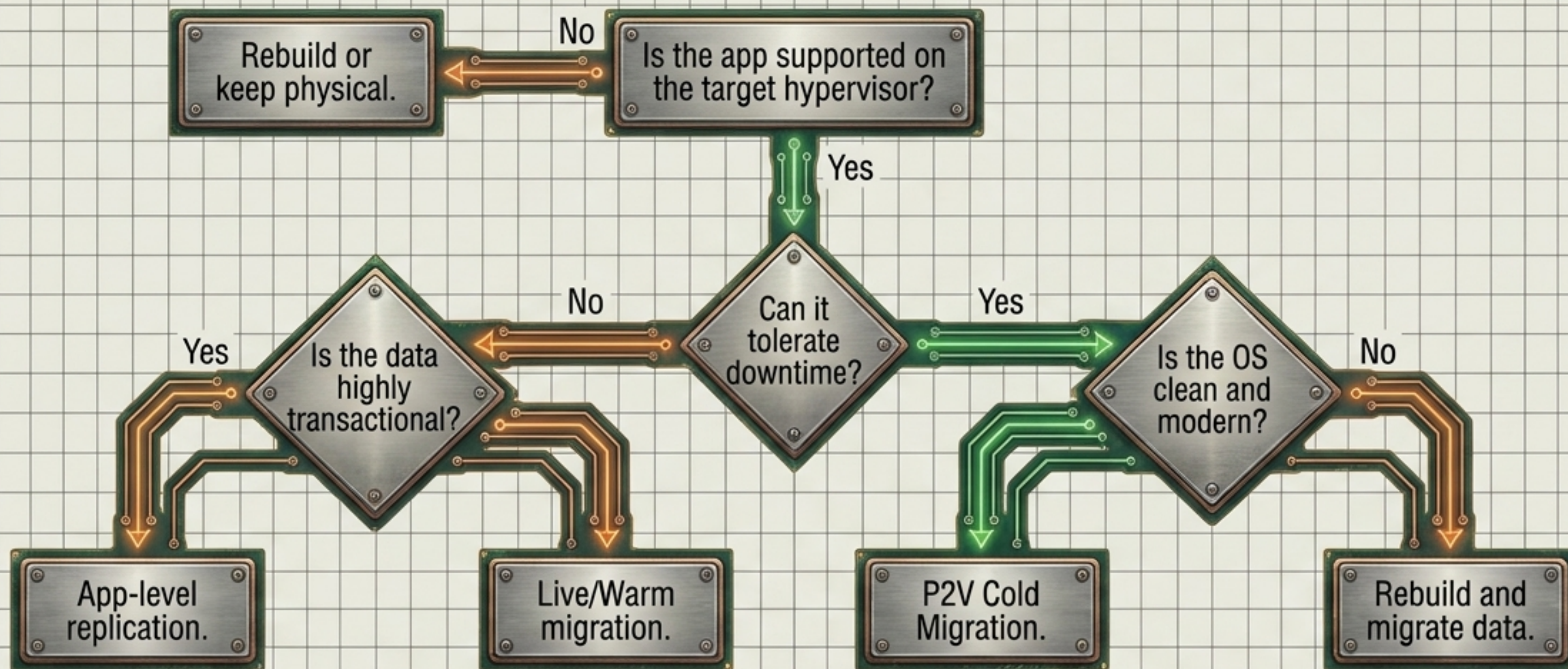
Never destroy the physical server immediately. Keep the source environment available until the observation period proves the workload is stable under real traffic.

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Top 5 Migration Killers

Risk	Impact	Mitigation
Storage Bottlenecks	VM has TBs of capacity but insufficient IOPS.	Baseline IOPS before move.
Network Identity Shifts	Linux renames physical NICs, MACs change, breaking IPs.	Post-conversion network validation.
Time Synchronization	Clock drift breaks Kerberos, TLS, and databases.	Unified NTP strategy.
Security & Backup Agents	Agents looking for physical UUIDs fail silently.	Test agent health post-cutover.
DNS Caching	TTL limits prevent new IP visibility.	Lower TTL before cutover.

The Engineer's Migration Decision Tree



A beginner asks: "How do I convert this server?"
An engineer asks: "What is the workload, what does it depend on, and how will I recover if it fails?"

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The Professional Mindset

1. Measure first.
2. Understand dependencies.
3. Plan the target.
4. Protect the rollback path.
5. Validate the business service.
6. Only then commit the change.

Migration is complete only when the workload is operational, secure, performant, backed up, and accepted by the business.

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